Practice Set 1 Paper 4 – Recent Advances

Contents

1. Q1. Briefly outline flexible endourological instrument and its role in management of urological disease (LAQ)
2. Q2. Describe energy sources for breaking stones in Urology (LAQ)
3. Q3a. Antioxidants in urological malignancy (SAQ)
4. Q3b. Bipolar energy (SAQ)
5. Q3c. Thalium laser in urology (SAQ)
6. Q3d. Morcellation (SAQ)
7. Q3e. KTP*** laser in urology (SAQ)
8. Q3f. Stem cell therapy (SAQ)

Practice paper compiled from the question bank.

Q1. Briefly outline flexible endourological instrument and its role in management of urological disease

Type: LAQ (25 marks) (flexible ureteroscopy in detail: see Practice Set 1 Paper 2, Q3b; digital scopes also in 2019-Summer Paper 4, Q3c)

Introduction

- A flexible endoscope has a **deflectable tip** and carries its image by a fibre bundle or a tip-mounted chip. It follows the natural curves of the urethra, ureter and collecting system.
- Physical basis: light travels along a curved glass fibre by **total internal reflection** (Tyndall).
- Marshall reported the first flexible URS (ureteroscopy) in 1964. It became therapeutic only when a **working channel** and **active tip deflection** were added in the early 1980s.

Components

Component	Key features
Optics	**Fibreoptic:** a *coherent* image bundle carries the picture; a *random* light bundle carries illumination; cladding improves reflection. **Digital:** CCD (charge-coupled device) or CMOS (complementary metal oxide semiconductor) chip at the tip with LED (light-emitting diode) light; no camera head or light cable
Deflection	Lever → control wires through movable metal rings → wire fixed at the tip. "Intuitive" = tip moves with the lever. The reticle (notch) marks the plane of deflection
Working channel	Single channel shared by irrigation and instruments, so instruments reduce flow and deflection
Shaft	Tapered tip; passive (secondary) deflection segment behind the tip helps lower-pole access

Types

Instrument	Specifications	Main use
Flexible cystoscope	14–17 Fr; 37–40 cm; deflection 120–220°; channel 6.4–7.5 Fr; field about 120°	Office diagnosis and minor therapy
Flexible ureteroscope	Tip 4.5–7.5 Fr; deflection ≥270° both ways; channel **3.6 Fr**; dual-channel scope (2 × 3.3 Fr) exists	Kidney and upper ureter
Flexible nephroscope (or flexible cystoscope used through the tract)	Larger calibre; too wide for the UPJ (ureteropelvic junction)	Clearing calyces during PCNL (percutaneous nephrolithotomy)

Fibreoptic vs digital

	Fibreoptic	Digital
Image	Honeycomb pattern; black dots as fibres break	Sharper, larger image; no focusing or white balance
Size	Smallest tips	Larger; more access-sheath use
Outcome	Reference	Shorter operating time, equal stone-free rate
Durability	Fragile; over-rotation damages fibres	Not clearly better in trials

- Surprise finding: in vitro, illumination was better with the fibreoptic cystoscope.

Single-use scopes

- **LithoVue (2016):** first single-use digital ureteroscope; CMOS, tip 7.7 Fr, shaft 9.5 Fr, 3.6 Fr channel, 270° each way. Single-use cystoscopes followed (2020).
- Rationale: reusable ureteroscopes may need repair after only **3–5 cases**; reprocessing and repair add cost.
- Cost advantage depends on case volume; environmental impact is similar to reusables.

Accessories

- **Guidewires:** 0.035/0.038 inch PTFE (polytetrafluoroethylene)-coated safety wire; hydrophilic wires to pass impacted stones (exchange, as too slippery for safety); nitinol-core and hybrid wires.
- **Access sheath** (11–16 Fr): repeated passes, lower intrarenal pressure; small risk of ureteric injury.
- **Laser fibres** 200–365 µm; **nitinol tipless baskets** ≤3 Fr; three-prong graspers; 3 Fr cup biopsy forceps; 3 Fr balloon dilating to 12 Fr; fulguration electrodes; pressure irrigation pumps.
- Flexible EHL (electrohydraulic lithotripsy) probes work; flexible pneumatic probes restrict deflection.

Role in urological disease

Area	Role
Lower tract	Haematuria work-up; bladder cancer surveillance, office biopsy and fulguration; NBI (narrow-band imaging) and blue light; stent removal or exchange; bladder stones, foreign bodies, strictures; botulinum toxin injection; diversions through a catheterisable channel; difficult ureteric access via a suprapubic tract
Stones	RIRS (retrograde intrarenal surgery): renal stones <20 mm, lower pole <10 mm; >2 cm when PCNL is unsafe
UTUC (upper tract urothelial carcinoma)	Diagnostic URS and biopsy (predicts grade in about 9 of 10); laser ablation of low-grade tumours; prefer to percutaneous route (seeding)
Obstruction	Endopyelotomy without crossing vessels; endoureterotomy for strictures <1.5 cm
Diagnosis	Positive cytology, filling defects, lateralising haematuria (haemangioma fulgurated), recurrent one-sided infection; foreign body retrieval
PCNL	Antegrade or retrograde flexible endoscopy raises stone-free rate; ECIRS (endoscopic combined intrarenal surgery)

Care and complications

- **Scope damage:** deflection wear; laser fibre scraping and perforating the channel. Pass fibres with the scope straight; fire only with the fibre tip seen on the stone; leave the bladder empty to avoid buckling.
- **Sterilisation:** ethylene oxide, glutaraldehyde or peracetic acid.
- **Patient:** ureteric injury, stricture, sepsis from high pressure. Use saline irrigation and antibiotic prophylaxis for all upper tract procedures.

Recent advances

- Pressure (and temperature) sensors in the scope; AI (artificial intelligence); robotic flexible URS; suction access sheaths; miniaturisation.

Diagram to draw: cross-section of a flexible ureteroscope: chip, LED, working channel and deflection wires.

Key points to remember:

- Working channel plus active deflection made flexible scopes therapeutic.
- fURS: 3.6 Fr channel, ≥270° deflection; cystoscope 14–17 Fr.
- Digital: better image, larger tip, durability not better.
- Golden rule: never fire unless the fibre tip is seen on the stone.

Q2. Describe energy sources for breaking stones in Urology

Type: LAQ (25 marks) (also covered in 2025-Winter Paper 4, Q2 and Q3b)

Introduction

- Stones are **brittle**: they crack under tension and shear. Tissue is **elastic** and survives the same stress. Every lithotripter exploits this difference.
- Energy reaches the stone **extracorporeally** (SWL, shock wave lithotripsy) or **intracorporeally** through an endoscope.

Classification

Group	Sources
Extracorporeal	SWL: electrohydraulic, electromagnetic, piezoelectric; BWL (burst wave lithotripsy, investigational)
Intracorporeal mechanical	Ballistic (pneumatic), ultrasonic, combined ultrasonic–pneumatic
Intracorporeal electrical	EHL (electrohydraulic lithotripsy)
Intracorporeal light	Lasers: Ho:YAG (holmium:yttrium-aluminium-garnet), TFL (thulium fibre laser), pulsed Tm:YAG (thulium:YAG); historic pulsed dye and FREDDY

1. Shock wave lithotripsy

- **Shock wave:** peak positive pressure 30–100 MPa, then a tensile tail of about −10 MPa. It is generated at F1, focused, and delivered to the stone at F2.

Generator	Mechanism	Strength	Weakness
Electrohydraulic (original HM3, 1980)	Underwater spark at F1 of an ellipsoid reflector	Very effective	Electrode life <40,000 shocks; focus drifts
Electromagnetic (modern standard)	Coil pushes a metal membrane; lens or reflector focuses	Consistent; high energy at F2; long life	—
Piezoelectric	Many ceramic elements on a spherical dish	Long life; little pain	Small focal zone, low energy

- **Fracture mechanisms:** shear and tear forces; **spallation** at the distal face; quasistatic squeezing; **cavitation** (bubble collapse erodes the surface); dynamic squeezing (most powerful). A focal zone wider than the stone works best.
- **Technique:** 60–90 shocks/min beats 120/min, because bubble clouds shield the stone and injure the kidney; voltage ramping; bubble-free gel coupling.
- Poor results: calcium oxalate monohydrate, brushite, cystine, density >1000 HU (Hounsfield units), skin-to-stone distance >10 cm.

2. Ballistic (pneumatic) lithotripsy

- Compressed air drives a projectile against a metal probe, a **jackhammer** (about 12 Hz).
- Breaks all stones cheaply with little ureteric injury. Drawbacks: **retropulsion**; rigid probes only, as flexible probes block deflection.

3. Ultrasonic lithotripsy

- A piezoceramic transducer vibrates a hollow probe at ultrasonic frequency; contact drills the stone.

- **Suction** through the hollow probe removes fragments at once. Best for soft stones in PCNL (percutaneous nephrolithotomy); weak on hard stones; needs irrigation for cooling.

4. Combined devices

- Ultrasonic plus ballistic in one probe (for example Trilogy, ShockPulse-SE): ballistic cracks hard stones while ultrasound and suction clear fragments. Faster PCNL.

5. Electrohydraulic lithotripsy

- A spark at the electrode tip forms a **vapour bubble**; expansion and collapse send pressure waves in all directions.
- Cheap; 3–5 Fr flexible probes. Poorly focused, so it injures the ureter and the scope; weak on cystine. AUA (American Urological Association) advises against it as first-line ureteric lithotripsy; still used for bladder stones.

6. Lasers

- **Pulse duration decides mechanism.** Short pulses (<10 µs: pulsed dye, FREDDY) are photomechanical, with cavitation shocks. Long pulses (>250 µs: holmium, thulium) are **photothermal**: the stone is heated until it melts, decomposes or flashes its water to steam.

	Ho:YAG	TFL
Wavelength	**2120 nm** (often quoted 2100)	1940 nm (water absorption peak)
Frequency	Up to about 120 Hz	Up to about 2000 Hz
Peak power	High	Low → less retropulsion
Fibre	≥200 µm	50–150 µm (more flow and deflection)
Product	Fragments and dust	Finer dust

- **Holmium:** standard since the 1990s; breaks **every** composition; flexible quartz fibres. Fragmenting 0.6–1.0 J × 6–10 Hz; dusting 0.2–0.3 J × 40–80 Hz; "popcorning" in a calyx.
- **Moses effect:** the start of the pulse boils a vapour channel through which the rest reaches the stone; pulse modulation reduces retropulsion.
- **TFL RCTs (randomised controlled trials):** superior to low-power (30 W) holmium; equal stone-free rate to 120 W pulse-modulated holmium.
- **Retropulsion** rises with pulse energy, not frequency, and falls with longer pulses.

Safety

- Laser heat follows total power: ≤10 W in the ureter, pauses, high flow, access sheath. Eyewear for all.
- Never fire without seeing the fibre tip on the stone.
- SWL: haematoma, arrhythmia, steinstrasse. Contraindications: pregnancy, coagulopathy or anticoagulants, untreated infection, aneurysm, distal obstruction.

Choice by site

Site	Usual energy
Ureter	Laser (semirigid or flexible); pneumatic distally; SWL
Kidney	SWL; laser via flexible URS (ureteroscopy); PCNL with ultrasonic, combined or laser
Bladder	Laser, pneumatic, EHL, mechanical

Recent advances

- **BWL:** handheld ultrasound bursts (5–100 cycles) break stones without anaesthesia. **Ultrasonic propulsion** moves fragments.
- Suction sheaths; AI (artificial intelligence)-adjusted laser settings.

Diagram to draw: SWL electrohydraulic ellipsoid with F1 and F2, stone at F2.

Key points to remember:

- SWL: electromagnetic is the modern standard; slow rate (60–90/min) is better.
- Holmium 2120 nm, photothermal, breaks every composition.
- TFL 1940 nm: low peak power, finer dust, less retropulsion.
- EHL is poorly focused, so it is not first-line in the ureter.

Q3a. Antioxidants in urological malignancy

Type: SAQ (10 marks) (prevention trials also in 2017-Summer Paper 4, Q3a)

Rationale

- ROS (reactive oxygen species) cause **oxidative DNA damage**; polyunsaturated fat and excess iron increase it. Antioxidants scavenge ROS, so diet studies suggested protection.
- Randomised trials of single high-dose supplements have been **null or harmful**.

Prostate cancer

Agent	Evidence	Result
SELECT (Selenium and Vitamin E Cancer Prevention Trial)	35,533 men; selenium 200 µg, vitamin E 400 IU, both, or placebo	Stopped early (median 5.46 years) for futility. No benefit. Vitamin E alone trended to more cancer (P = 0.06); extended follow-up showed a significant **increase (about 17%)**
ATBC (Alpha-Tocopherol, Beta-Carotene)	Low-dose vitamin E in heavy smokers	Secondary analysis: prostate cancer ↓32%, deaths ↓41%; beta-carotene ↑ lung cancer
Selenised yeast	Skin-cancer trial	Secondary fall in prostate cancer prompted SELECT
Lycopene (cooked tomato)	Meta-analysis: modestly lower risk, mainly advanced cancer	WCRF (World Cancer Research Fund) downgraded "probable" (2007) to "no conclusion" (2018); supplement trials mostly null
Green tea catechins	HGPIN (high-grade prostatic intraepithelial neoplasia)/ASAP trials	Large fall in progression to cancer (meta-analysis); dose impractical
Vitamin C, high-dose antioxidants	High doses act as pro-oxidants	Supplements above median → more advanced cancer

- **Also know:** vitamin D (VITAL, no effect); coffee (modestly lower advanced risk); pomegranate, soy and curcumin unproven.

Bladder cancer

- Supplemental beta-carotene, vitamins C and E and folate do not lower risk; SELECT showed no effect.
- Fruit and vegetable benefit disappears after adjusting for smoking. Smoking cessation is the real prevention.

Kidney cancer

- No antioxidant trial has shown benefit.

Why the trials failed

- Single nutrient at high dose; wrong chemical form; non-smoking populations; PSA-era indolent cancers; U-shaped dose effects.

Current position

- Antioxidant supplements are **not recommended** for cancer prevention. Advise a balanced diet, no smoking and normal weight.

Diagram to draw: ROS → DNA damage → cancer, with antioxidants blocking the first arrow.

Key points to remember:

- SELECT: no benefit; vitamin E alone increased prostate cancer.
- ATBC benefit limited to smokers on low dose.
- Lycopene downgraded to "no conclusion" (2018).

Q3b. Bipolar energy

Type: SAQ (10 marks) (also covered in 2025-Winter Paper 4, Q2)

Principle

- Radiofrequency current (about 500 kHz) passes **only between two electrodes on the same instrument**. No dispersive pad; the patient is not in the circuit.
- Less power and less lateral thermal spread (1–6 mm vs 5–7 mm monopolar). In endoscopy, **saline** irrigation can be used.

Monopolar vs bipolar

	Monopolar	Bipolar
Circuit	Loop → patient → pad	Active and return poles on the instrument
Irrigant	Glycine 1.5%, sorbitol–mannitol	Normal saline
TUR (transurethral resection) syndrome	Yes	Theoretically eliminated
Capacitive coupling, pad burns	Yes	No

Endoscopic uses

- **Bipolar TURP (transurethral resection of prostate):**
  - *Plasma system:* energy vaporises saline around the loop into plasma (orange glow of sodium); cuts at lower temperature while sealing vessels; platinum–iridium loop.
  - *TURis (transurethral resection in saline):* the sheath acts as return electrode; chapter notes report strictures and fistulae.
  - *True bipolar:* dual loops.
  - Outcomes: equal IPSS (International Prostate Symptom Score) and Qmax; less transfusion and clot retention; shorter catheter time and stay. NNT (number needed to treat) about 50 to prevent one TUR syndrome.
- **Bipolar TUVP (transurethral vaporisation):** PlasmaButton; AUA (American Urological Association) option for <30 and 30–80 cc.
- **Bipolar enucleation:** plasmakinetic energy can perform AEEP (anatomic endoscopic enucleation of prostate) satisfactorily; comparative data limited.
- **TURBT (transurethral resection of bladder tumour):** fewer obturator jerks and less bleeding in reviews; one RCT (randomised controlled trial) showed only less thermal artefact.

Laparoscopic and open uses

- Bipolar forceps for pinpoint haemostasis.
- **Advanced vessel sealers (LigaSure, EnSeal, Caiman):** compression plus bipolar current fuses collagen and elastin. Impedance feedback stops output. Seal vessels **≤7 mm**; spread 1–3 mm; one application only.
- Uses: prostatic pedicles, cystectomy.

Safety

- Safer near pacemakers; touching tines complete the circuit without tissue.
- Obturator reflex reduced, not abolished.

Diagram to draw: monopolar loop-to-pad circuit beside bipolar loop with local return.

Key points to remember:

- Bipolar: no pad, saline irrigation, no TUR syndrome.
- Bipolar TURP equals monopolar efficacy with fewer complications.
- Vessel sealers: vessels ≤7 mm, spread 1–3 mm.

Q3c. Thalium laser in urology

Type: SAQ (10 marks) (also covered in 2024-Winter Paper 4, Q3c)

Assumed meaning: thulium laser ("thalium" is a spelling error).

Types and physics

	Tm:YAG (thulium:yttrium-aluminium-garnet)	TFL (thulium fibre laser)
Wavelength	2013 nm	**1940 nm** (water absorption peak)
Source	Diode-pumped solid crystal	Diode-pumped thulium-doped silica fibre
Output	Continuous wave (now also pulsed)	Pulsed or continuous
Water absorption	High	About 4× holmium

- Optical penetration is shallowest of urological lasers: about 0.2 mm (chapter values for TFL: 0.08–0.2 mm), against 0.31–0.4 mm for Ho:YAG (holmium:YAG, 2120 nm).
- Shallow absorption gives precise vaporisation and clean incision, but **carbonisation**.

Stone lithotripsy (TFL)

- Frequency up to about 2000 Hz (holmium about 120 Hz); pulse energy as low as 0.025 J.
- **Low peak power, long pulse** → photothermal "dusting", minimal retropulsion.
- Fibres 50–150 µm → better irrigation and scope deflection. Small, quiet unit; no dedicated high-voltage outlet.
- Settings: start 0.6 J × 6 Hz; dusting 0.2 J × 40 Hz; avoid >40 Hz (pulses miss the stone). "Paint, do not drill."
- RCTs (randomised controlled trials): better than 30 W holmium; equal stone-free rate to 120 W pulse-modulated holmium with less retropulsion.

Prostate

- **Techniques:** ThuVARP (vaporesection), ThuVAP (vaporisation), ThuVEP (vapoenucleation), **ThuLEP** (enucleation, 2010; mostly blunt dissection), ThuFLEP (fibre laser enucleation).
- Settings: 70–90 W enucleation, 30 W coagulation; morcellation as for HoLEP (holmium laser enucleation).
- Outcomes: ThuLEP equals HoLEP; size-independent; safe on anticoagulants (AUA [American Urological Association]). Retrograde ejaculation about four-fifths.

Other uses

- **Also know:** en bloc bladder tumour resection; UTUC (upper tract urothelial carcinoma) ablation; incision of strictures.

Safety

- Heat follows power: ≤10 W in the ureter, high flow, access sheath. Eyewear for all.
- Concerns: efficacy on all stone types; thermal injury at high power.

Diagram to draw: water absorption curve marking 1940, 2013 and 2120 nm.

Key points to remember:

- TFL 1940 nm, Tm:YAG 2013 nm; water absorption about 4× holmium.
- TFL: low peak power, fine dust, thin fibres.
- ThuLEP is size-independent and equal to HoLEP.

Q3d. Morcellation

Type: SAQ (10 marks)

Definition

- Cutting a large specimen into small pieces so it leaves through a port, small incision or endoscope.
- **Manual:** ring forceps or Kelly clamp inside a bag. **Power (electromechanical):** rotating cylindrical blade (laparoscopy) or endoscopic tissue morcellator (prostate).

1. Endoscopic morcellation after enucleation

- Made **HoLEP (holmium laser enucleation of prostate)** practical; also ThuLEP (thulium laser enucleation).
- **Steps:**
  1. Achieve good haemostasis first.
  2. Pass a long nephroscope with a 5 mm channel.
  3. Run two inflow lines fully open to keep the bladder distended.
  4. Engage lobes with suction; cut only in mid-bladder, away from the wall.
  5. Collect fragments in a trap for histology.
- Devices: Piranha (rotating blades), VersaCut (oscillating).
- **Complications:** bladder mucosal injury (stop, release suction, disengage; mostly catheter only); intraperitoneal perforation needs laparotomy; **retained adenoma** (from malfunction or poor vision) needs repeat morcellation.

2. Laparoscopic morcellation (kidney, spleen)

- **LapSac** (nylon with polyurethane lining) is impermeable to tumour cells and bacteria.
- **Rules:** purpose-built bag; drapes; change gloves and instruments after; no deep forceps passes; keep pneumoperitoneum and watch the bag.

	Intact extraction	Morcellation
Incision	Larger	Small, better cosmesis
Pain, time, stay	Similar	Similar
Pathology	Full staging	Type and grade kept; size, capsule, vein invasion, and depth in UTUC (upper tract urothelial carcinoma) lost
Seeding	—	Rare port-site recurrence

- Splenectomy morcellation risks **splenosis** if the bag ruptures.

3. FDA 2014 warning (gynaecology)

- In 2014 the FDA (Food and Drug Administration) warned against laparoscopic power morcellation in hysterectomy or myomectomy for fibroids, because unsuspected uterine sarcoma (about 1 in 350) was spread and upstaged.
- A boxed warning followed; contained (in-bag) morcellation is advised.
- Lesson for urology: avoid uncontained power morcellation of possible malignancy.

Complications summary

- Bag rupture, seeding, lost staging, visceral or vascular injury, retained tissue.

Diagram to draw: enucleated lobes in a distended bladder with morcellator suction.

Key points to remember:

- Morcellator made HoLEP practical; distend bladder, haemostasis first.
- Morcellation loses stage information; histology still obtained.
- FDA 2014: uncontained power morcellation spreads occult sarcoma.

Q3e. KTP*** laser in urology

Type: SAQ (10 marks) (lasers in BPH also in 2017-Winter Paper 2, Q3a)

Assumed meaning: KTP (potassium titanyl phosphate) laser, the "GreenLight" laser; the asterisks are ignored.

Physics

- A 1064 nm Nd:YAG (neodymium:yttrium-aluminium-garnet) beam passes through a **KTP or LBO (lithium triborate) crystal**. This doubles frequency and halves wavelength to **532 nm** (green).
- Green light is absorbed by **haemoglobin**, not water. Optical penetration 0.8 mm; tissue effect 1–3 mm.
- Effect: **photoselective vaporisation** with a thin coagulation rim, so vaporisation is bloodless.

Platforms

Generation	Power
KTP	80 W
LBO HPS (High Performance System)	120 W
LBO XPS	**180 W**; 750 µm MoXy fibre (metal tip, fluid cooling)

- Side-firing fibre emits at 70°; 23–24 Fr cystoscope; saline irrigation.

PVP (photoselective vaporisation of prostate) technique

- Identify ureteric orifices; open the bladder neck first.
- Sweep 15–30° in 1–2 seconds at ≤2 mm working distance. Large bubbles mean efficient vaporisation.
- Pulsatile bleeding: coagulation mode (about 30 W). Energy >5 kJ/cc predicts complete vaporisation.

Indications

- AUA (American Urological Association): prostates <30 cc and 30–80 cc; men at **higher bleeding risk** (anticoagulation continued). Not >80 cc (high retreatment).

GOLIATH trial

- RCT (randomised controlled trial), 29 European centres, about 270 men, prostate ≤100 cc: **180 W PVP vs TURP** (transurethral resection of prostate).
- PVP **non-inferior** for IPSS (International Prostate Symptom Score), Qmax and complications at 6 months, 1 and 2 years.
- PVP: shorter catheter time and stay, fewer early reinterventions. TURP: faster. Retrograde ejaculation (about two-thirds) and 2-year reoperation equal.

Complications

- Dysuria and storage symptoms (highest of transurethral techniques); UTI (urinary tract infection) and epididymitis from coagulum; capsular perforation 0.2–1%; retrograde ejaculation.
- No TUR (transurethral resection) syndrome; transfusion very rare.
- **No tissue** for histology.

Other uses

- **Also know:** laparoscopic non-contact cutting (largely supplanted); KTP enucleation possible.

Diagram to draw: side-firing fibre at 70° vaporising lateral lobe.

> **Key points to remember:**
> - 532 nm: frequency-doubled Nd:YAG; haemoglobin chromophore.
> - GOLIATH: 180 W PVP non-inferior to TURP.
> - Best for anticoagulated men; not >80 cc.

Q3f. Stem cell therapy

Type: SAQ (10 marks) (tissue engineering also in 2023-Winter Paper 1, Q1)

Definition

- Stem cells **self-renew** and **differentiate** into specialised cells. Therapy uses them to repair or replace damaged tissue.

Types

Type	Source	Comment
Embryonic	Blastocyst	Pluripotent; ethics, teratoma
Adult mesenchymal	Bone marrow, adipose (ADSC, adipose-derived stem cells), umbilical cord	Easy harvest; mainly paracrine
Muscle-derived (MDSC)	Skeletal muscle biopsy	Used for sphincter and detrusor
iPSCs (induced pluripotent stem cells)	Reprogrammed adult cells	Patient-specific; tumour risk
Urine-derived, spermatogonial	Urine; testis	Research

Mechanisms

- Differentiation into muscle, nerve or endothelium.
- **Paracrine action:** growth factors, anti-inflammatory, anti-fibrotic and neurotrophic signals.
- Homing to injured tissue; seeding of scaffolds.

Applications

- **SUI (stress urinary incontinence):**
  - Autologous muscle-derived cells: thigh muscle biopsy → culture → transurethral or periurethral injection into the sphincter.
  - Aim: regenerate the striated sphincter, not just bulk it.
  - Early studies improved; placebo-controlled trials have not shown clear superiority, so it remains investigational.
  - After childbirth injury, stem cell homing chemokines (CXCL12, CCL7) rise.
- **ED (erectile dysfunction):** intracavernosal bone-marrow, adipose or cord cells; animal and small uncontrolled human series only. SMSNA (Sexual Medicine Society of North America): experimental, IRB (institutional review board) protocols only.
- **Underactive detrusor:** muscle-derived cells injected into the bladder wall; two open-label studies improved voiding; research.
- **Tissue engineering:** cell-seeded scaffolds for bladder (Atala, 2006) and urethra (Raya-Rivera, 2011); adipose stem cell-seeded grafts for Peyronie disease (experimental).
- **Fertility:** spermatogonial stem cell banking in prepubertal cancer patients.
- **Kidney:** iPSC kidney organoids (research).

Challenges

- Poor cell survival and retention; tumour formation (embryonic, iPSC); standardisation, cost and regulation.
- Unregulated "stem cell clinics" offer unproven treatments outside trials, a growing ethical concern.

- **Also know:** cancer stem cells (a different concept) may drive tumour recurrence and resistance.

Diagram to draw: muscle biopsy → culture → periurethral injection into sphincter.

> **Key points to remember:**
> - Adult stem cells act mainly by paracrine signals.
> - SUI muscle-derived cells and ED stem cells are investigational.
> - Embryonic and iPSCs carry tumour risk.

Rapid Revision Sheet – Practice Set 1 Paper 4

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- TFL 1940 nm: low peak power, finer dust, less retropulsion.
- EHL is poorly focused, so it is not first-line in the ureter.

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- SELECT: no benefit; vitamin E alone increased prostate cancer.
- ATBC benefit limited to smokers on low dose.
- Lycopene downgraded to "no conclusion" (2018).

Q3b. Bipolar energy (SAQ)

- Bipolar: no pad, saline irrigation, no TUR syndrome.
- Bipolar TURP equals monopolar efficacy with fewer complications.
- Vessel sealers: vessels ≤7 mm, spread 1–3 mm.

Q3c. Thalium laser in urology (SAQ)

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